Prefill Once, Ask Many: Training-Free Reversible Below-Page KV Selection for Multi-Query Long-Document Visual Question Answering

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Abstract

Real document-VQA workloads are multi-query: one long document is prefilled once and asked many different questions. Standard KV-cache eviction commits to a single compressed context and reuses it for every question. We present PAQ-KV ("prefill once, ask many"), a training-free architecture that makes per-query re-selection over a *retained full-context* KV cache inexpensive for a frozen vision–language reader: the document is prefilled once into the reader's KV cache and kept intact; per query, the reader's own question-to-content attention scores 64-token, page-aligned KV blocks, and the answer is decoded under a restricted-attention 4D mask over the intact cache—*below-page masked decoding with no gather, so every M-RoPE position is the model's own*—requiring no re-read, no second model, and no index. Because nothing is discarded, selection is fully reversible—the property the architecture unlocks—and per-query re-selection beats the identical scorer committed once, a strong gold-informed static, and a trained retriever's committed form (isolation contrast $+0.264$, $\text{LB}_{95} = +0.187$ under a clean-cache re-run; block-level replication $+0.379$, $\text{LB}_{95} = +0.273$). At a nominal 5% (realized $\approx 6\%$) decode-attention budget, PAQ-KV beats ColQwen2, a state-of-the-art trained visual retriever, on MMLongBench-Doc (0.4042 vs. 0.3282 with ColQwen2's trained retrieval-projection LoRA correctly loaded; $+0.077$, document-clustered $\text{LB}_{95} = +0.040$; 80 documents, 444 decoded / 440 paired), matches full context, and is *not statistically distinguishable from* the gold-page oracle at its $\sim 6\%$ budget (-0.026, CI straddles zero—parity within noise, not a demonstrated equality). A wrong-question control collapses accuracy, supporting query-specificity, and below-page granularity outperforms page-level. On LongDocURL PAQ-KV matches but does not beat the retriever ($+0.019$, $\text{LB}_{95} = -0.019$; below full context by -0.014); we report this as a limitation and observe the cell appears largely reader-conversion-bound rather than selection-bound. Two selector upgrades (a per-head mixture transfer gate that stopped the LoRA lane before any adapter was trained; a sharp expert-head scorer) failed pre-registered gates and are reported as negatives. All results are development-pin evidence on a reused pin; a sealed held-out set is reserved for a single final evaluation.

1 Introduction

Long-context vision–language models (VLMs) can read an entire multi-page document, but they pay for every page on every question. The standard efficiency response, inherited from single-query text benchmarks, is to *evict*: keep a compact subset of the KV cache and decode against it (Li et al. 2024; Zhang et al. 2023; Chen et al. 2024; Wan et al. 2024; Kim et al. 2025). Eviction is a single, irreversible commitment: whatever subset is kept must serve every future question about that document.

Document assistants, however, are not asked one question. The two long-document VQA benchmarks we use are natively multi-query: grouping their 64K splits by source document, MMLongBench-Doc (Ma et al. 2024) averages ≈ 6.97 questions per document and LongDocURL (Deng et al. 2025) ≈ 3.6; inside our pinned evaluation subsets the averages are 5.50 and 5.79 gold-bearing questions per document. Different questions need different evidence, so a subset committed for question 1 is *stale* for question 5.

This paper asks: at a matched budget on long multimodal documents, does per-query *re-selection* of context beat a *commit-once* selection—and can a training-free, reader-internal selector compete with a dedicated trained retriever? Our answer is PAQ-KV: prefill the full document *once* into the KV cache of a frozen reader (Qwen3-VL-8B; Qwen Team 2025) and retain it; per query, re-select the most relevant 64-token, below-page KV blocks with the reader's own question$\rightarrow$content attention; decode under a restricted-attention 4D mask over the retained full-context cache. Nothing is discarded and nothing is gathered (M-RoPE positions are untouched), so a selection is a *view*, not a commitment: the next question re-selects from the same intact cache.

Contributions.

- **An architecture for cheap per-query re-selection over a retained cache.** PAQ-KV keeps the full-document prefill intact and, per query, re-selects 64-token below-page KV blocks and decodes under a 4D restricted-attention mask *without gathering* the cache, so all M-RoPE positions stay the model's own. This is what makes per-query re-selection cheap and exact: no re-read, no second model, no index, one deployed artifact. An identity audit confirms the mask decode is identical to the native decode in all audited cases.

- **Reversibility, the property the architecture unlocks.** Because nothing is discarded, a selection is a view rather than a commitment. Under a pre-registered two-comparator gate, per-query re-selection beats the identical scorer committed once ($+0.257$/$+0.163$ per cell; 903 queries, 80 documents/cell, document-clustered) *and* a strong gold-informed static ($+0.108$/$+0.104$); the contrast survives a clean-cache re-run ($+0.264$, $\mathrm{LB}_{95} = +0.187$), reappears below page level ($+0.379$, $\mathrm{LB}_{95} = +0.273$), and also holds for a trained retriever's committed form—so the effect is a property of the amortized regime the architecture serves, not merely fresh-beats-deliberately-stale for one scorer.
- **A powered, adversarially audited beat over a trained retriever.** On MMLongBench-Doc, PAQ-KV at a nominal 5% (realized $\approx$6%) budget scores 0.4042 vs. ColQwen2 0.3282 ($+0.077$, $\mathrm{LB}_{95} = +0.040$; 80 documents, 444 decoded / 440 paired; ColQwen2 evaluated with its trained retrieval-projection LoRA correctly loaded, §4.4), is indistinguishable from full context, and is not statistically distinguishable from the gold-page oracle at its $\sim$6% budget (-0.026, CI straddles zero). The audit shows selection is query-dependent, query-specific (a wrong-question control collapses accuracy, supporting query-specificity), reversible, and finer than pages by a significant margin.
- **An honest limitation and two gated negatives.** On LongDocURL PAQ-KV matches but does not beat ColQwen2 ($+0.019$, $\mathrm{LB}_{95} = -0.019$; below full context by -0.014); §6 characterizes it as an apparently conversion-bound cell. A per-head mixture transfer gate (which stopped the LoRA lane before any adapter was trained) and a sharp expert-head block scorer both failed pre-registered gates and are reported at full prominence.

Evidence status. Every number in this paper is a development-pin result on an 80-document-per-cell pin; a sealed, disjoint held-out set (73/45 documents) is reserved for a single one-touch final evaluation that has not been run. We state this scope wherever it binds.

2 Related Work

Page-level attention selection. CAPS (Zhu et al. 2026) publishes attention-based *page* selection on our exact benchmarks: a scoring VLM's final-token$\rightarrow$page attention at one expert (layer, head), lightly calibrated off-benchmark, beats ColPali-pipeline retrieval. We state plainly that the "reader-attention selects pages" method is CAPS's. PAQ-KV differs on the axes CAPS itself names as open: it operates *below* page granularity, it is *reversible* over one retained prefill rather than re-forwarding every page per query (CAPS reports 11.25 s/query and concedes the multi-query regime to embedding methods), and it decodes directly from the masked cache with no re-read.

Visual document retrieval. ColPali/ColQwen2 (Faysse et al. 2025) embed pages once and re-rank per query; M3DocRAG (Cho et al. 2024) and successors build pipelines on such retrievers; page-selection-then-read itself long predates this line (Tito, Karatzas, and Valveny 2024). A published oracle analysis on MMLongBench (Anonymous 2026b) mirrors our own complementarity findings. ColQwen2 is the strong deployable baseline we compare against throughout.

KV compression and eviction. SnapKV (Li et al. 2024), H2O (Zhang et al. 2023), FastV (Chen et al. 2024), LOOK-M (Wan et al. 2024), and KVzip (Kim et al. 2025) commit a compressed cache once. Quest (Tang et al. 2024) shows query-aware selection beats query-agnostic eviction at tight budgets in single-query text—motivation for, but not evidence of, our multi-query claims. SparseVILA (Anonymous 2025d) combines prefill pruning with decode-time retrieval but prunes irreversibly; Kamera (Anonymous 2026c) shows the multi-query KV-reuse serving regime is becoming practically real.

Region-level and attention-guided neighborhood. Region- and crop-level document methods are active: RegionRAG (Yuan et al. 2025) (trained retriever-side crops), Snappy (Anonymous 2025c) (patch-to-OCR-box localization), AGREE (Anonymous 2025a) (reader attention distilled into a trained retriever), agentic crop/zoom loops (Wang et al. 2025; Anonymous 2026a), and training-free single-image cropping (Zhang et al. 2025; Anonymous 2025b); attention-based reranking precedents exist in text (Chen, Gutierrez, and Su 2025; Wu et al. 2025), and MM-DocIR (Dong et al. 2025) supplies layout-level labels. Our search did not identify prior work in the exact combined cell—reader-internal attention as the runtime below-page selector, end-to-end QA at matched budget against page-level retrieval, on long multi-query documents—which we offer as a provisional literature-map claim, not proof of novelty.

3 Method: PAQ-KV

3.1 Setting

One long visual document, consumed as page images (no OCR), is read by a frozen VLM and will be asked k questions (5.5–5.8 gold-bearing per document in our pins). The document is prefilled **once** at the same rendering the full-context arm uses ($\leq$64K tokens) and the KV cache is *retained*: never evicted, never physically compressed, never gathered. The cache plays the role a multi-vector index plays for a retriever—except it is the reader's own working memory, so retrieval and reading share one representation and one model.

3.2 Per Query: Preview, Re-Select, Masked Decode

1. **Preview (score).** The tokenized question is forwarded over the retained cache—with the cache crop-restored afterwards, so queries never contaminate one another (§4.4)—and the question rows' attention over context positions is aggregated into scores over **64-token, page-aligned KV blocks**: fine enough to isolate a table row, a figure caption, or a paragraph; coarse enough for the scores to be stable.

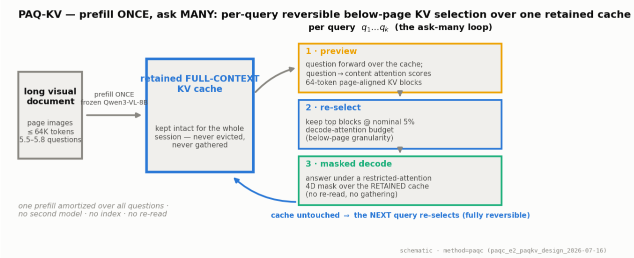

Figure 1: The PAQ-KV loop. The document is prefilled once into the frozen reader's KV cache, which is retained intact for the whole session. Each query runs the same three steps—attention preview, block re-selection, masked decode—against that one cache. No second model, no index, no full-resolution re-read; the cache is never modified, so selection is reversible by construction.

2. **Re-select.** Keep the top-scoring blocks at a **nominal 5% decode-attention budget**. The terminology is deliberate: the kept blocks' K/V were computed with full-document context during the prefill (the standard claim semantics of the Quest/SnapKV method class), so this is a decode-attention budget, not a matched-token prefill comparison; the realized attended fraction on the executed screen is $\approx$5.9–6.0% of context and is always reported. The matched quantity against ColQwen2 is the evidence budget (5% of pages vs. 5% of context) at identical reader and scorer.
3. **Masked decode.** The answer is generated while attending only to the selected blocks plus a small **always-keep set** (the BOS/system/template scaffold, the question itself, and previously generated tokens), enforced by a **4D additive attention mask** (batch, head, query row, key) over the retained cache. **No gathering**: the K/V tensors stay at their original positions, so every M-RoPE rotary index is exactly the model's own—the failure mode that makes physical cache-gathering treacherous for VLMs is sidestepped entirely. No re-read, no re-render, no second model.

Algorithm 1 gives the loop; Figure 1 shows it schematically and Figure 2 the mask machinery. Attention is captured by OOM-safe per-layer forward hooks that reduce each layer's attention tensor immediately and discard it, so peak memory is one layer's attention (the naive all-layers path OOMs at long contexts).

3.3 Exactness and Reversibility

A decode-identity audit checked the mask path against the native decode over 6 document–query pairs deliberately spanning 32K–122K tokens (a stress range wider than the $\leq$64K evaluation pin, to probe long-context behavior): in **all 6 audited cases** an all-true mask produced **zero mask-specific argmax changes** relative to the native causal decode (an earlier single greedy-token divergence traced to the generation loop's mask-independent fast-path kernel, not to the mask). We therefore claim the mask decode is identical to the native decode in the audited cases, not a general proof; this audit was a pre-registered kill criterion. Because nothing is discarded, a selection is a view, not a commitment: the next query re-scores and re-masks the same complete cache, whereas commit-once eviction cannot—what it dropped is gone. Reversibility is our tested axis (§4.2), and the contrast replicates at block level (+0.379, $\text{LB}_{95} = +0.273$; §5.2).

Algorithm 1 PAQ-KV: per-query reversible below-page KV-block selection with a 4D restricted-attention mask

Require: document pages rendered as in the full-context arm ($\leq$64K tokens); queries $q_1, \ldots, q_k$; nominal budget $b = 5\%$; frozen VLM
prefill the document once; retain the full KV cache C (context length n_{ctx}); record page-aligned 64-token block spans $B_1, \ldots, B_m$ and the always-keep scaffold set A
for each query q_j **do**
  forward the tokenized question over C; capture question→context attention via per-layer hooks; **crop** C back to n_{ctx}
  $\text{score}(B_i) \leftarrow$ aggregated question-row attention mass on block B_i's positions
  $S_j \leftarrow$ top blocks by score at the nominal budget $b \cdot n_{\text{ctx}}$
  build the 4D additive mask M_j: decode rows may attend to $A \cup S_j \cup q_j \cup$generated tokens; all other context positions get $-\infty$
  decode the answer from C under M_j—no gathering: K/V positions and M-RoPE indices unchanged
  discard M_j; C is intact for q_{j+1}
end for

What PAQ-KV is not. It is not CAPS (a separate scoring VLM, per-page full re-forwards, page granularity, single-query); not coarse-to-fine page routing with a full-resolution re-read (our own earlier page-level system, §5.5); and not eviction (the mask is per-query and reversible). The mask route is the *accuracy* path; physically gathering the cache for savings is a separate, future efficiency claim we do not make.

4 Experimental Setup

4.1 Benchmarks, Pins, Reader

We evaluate on **LongDocURL** (Deng et al. 2025) and **MMLongBench-Doc** (Ma et al. 2024), both as page images. For each document we build the union of the benchmark's own-document pages across that document's queries (dropping cross-document 64K padding), so every query's evidence is present in one prefill. Documents qualify with $\geq$ 3 gold-bearing queries and $\leq$ 110 union pages ($\approx$64K tokens at read resolution); selection is seed-0 frozen. The confirmatory pin holds 80 documents per cell (Table 1); a 30-document screen pin is its strict prefix, and results are additionally checked on the 50 new documents only (§5.1). The reader is **Qwen3-VL-8B** (Qwen Team 2025), frozen, eager attention. Page-level arms operate at a matched page budget $b \in \{2.5\%, 5\%, 10\%\}$ (the pre-registered gate budget is $b = 5\%$); accuracy is the benchmark's document-VQA scorer. Everything in this paper is a Qwen3-VL-8B

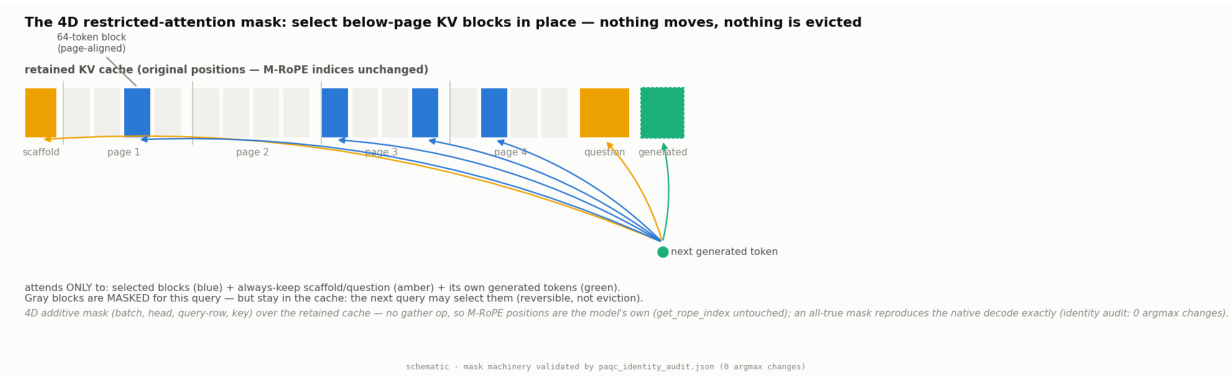

Figure 2: The mechanism. The retained cache is a strip of K/V at their original positions: scaffold, page content in 64-token page-aligned blocks, question, generated tokens. The 4D additive mask lets decode rows attend only to the per-query selected blocks plus the always-keep set; everything else is masked *for this query only*—it stays in the cache and remains selectable by the next query. Because nothing is gathered, the M-RoPE position machinery is untouched, and an all-true mask reproduced the native decode in every audited case (identity audit: zero mask-specific argmax changes over the audited pairs).

	LongDocURL	MMLongBench
Documents	80	80
Complete queries at gate	463	440
Gold-bearing queries / doc	5.79	5.50
Median (mean) union pages	44 (45.0)	28 (36.5)

Table 1: The confirmatory evaluation pin. "Complete queries" are rows where every gated arm scored; 4 of 907 vision queries are missing due to a per-document timeout, and worst-case imputation leaves every gate verdict unchanged.

case study; no second VLM has been run.

Base-model ceiling. A large fraction of queries receive no credit even given exactly the annotated gold pages: the oracle-evidence arm scores 0 on $317/903 = 35.1\%$ of confirmatory queries and a full 1 on only 45.1%. Pairing controls for shared query difficulty; the oracle is *not* a strict ceiling (a selected page set can beat the annotated gold set on individual queries).

4.2 Statistical Protocol

All inference uses a **document-cluster paired bootstrap**: queries of one document share the document and (for stale arms) the single committed subset, so bootstrapping over queries would be anti-conservative pseudoreplication. Intervals resample documents with replacement within each cell (10,000 draws, seed 0); LB_{95} denotes the 2.5th percentile of the bootstrap distribution—the lower end of a two-sided 95% CI. Per-cell results are primary; pooled contrasts use equal cell weighting and are labelled. Every gate in this program was frozen before its run.

The isolation gate. "Query-aware beats query-agnostic" is both a known text result (Tang et al. 2024) and a strawman risk. We therefore fix the tested axis to **reversibility**: every baseline runs its canonical scorer but is deployed *stale*—committed once per document (query-aware scorers commit a min-max-normalized aggregate over the document's first $M = \min(3, k)$ queries)—while the fresh arm re-selects per query. Reversibility is credited only if the fresh arm beats **both** (i) its *own* scorer committed once (identical scorer and budget, so the delta isolates the mechanism) and (ii) a strong *gold-informed* static subset (top pages by gold-page frequency across the document's queries—not deployable, and not provably optimal, but immune to the weak-static artifact that invalidated an earlier one-comparator gate).

4.3 Baselines

The executed panel (Appendix A) compares, per query at matched budget on the same pinned data and reader: **ColQwen2** (Faysse et al. 2025) fresh and stale, **CLIP** (Radford et al. 2021) fresh, SnapKV fresh/stale, the isolation arms above, random/truncation controls, and a demoted query-blind eviction family (H2O; a context-only FastV proxy; LOOK-M; a KVzip $\mathrm{recv\_max}$ proxy)—disclosed proxies that sit at or below random page recall here and are never claimed as beaten opponents. Full-context and gold-page-oracle rows are references, not competitors.

4.4 Methods Integrity

A cross-family build review found a real bug in the per-query scoring loop used across this program: the runtime mutates the retained cache in place, and the loop reused the document prefill across a document's queries without restoring it, so query q_i scored while attending to $q_0 \ldots q_{i-1}$. The fix (crop the cache after every per-query preview) is now applied everywhere, and its impact was quantified rather than assumed: on a 24-document/150-query clean-vs-polluted rerun, the key isolation contrast is $+0.264$ ($\text{LB}_{95} = +0.187$) clean vs. $+0.284$ polluted—the contrast largely cancels the bug because both arms share the same scorer—while absolute fresh-arm accuracy carries ≈ 0.04 inflation in the (page-level, pre-fix) confirmatory run, which we disclose wherever absolute levels matter. All PAQ-KV results postdate the fix.

Baseline-loader integrity. A second cross-family review caught a defect in the *comparator*, not our method: the cached ColQwen2 baseline had run with its trained final retrieval-projection LoRA (custom_text_proj) silently zeroed. vidore/colqwen2-v1.0 is a PEFT adapter, and under our peft/transformers versions a key-remap inserted language_model into the LoRA target path—correct for the language-model layers but wrong for the top-level custom_text_proj—so the checkpoint's projection-LoRA tensors landed at a non-matching path and the final projection ran with a zero LoRA delta (base weights only). Because the other adapter layers still loaded, the baseline was functional (well above random), which masked the defect until we inspected the load report and the projection weights directly. We patched the loader (copy the two checkpoint custom_text_proj LoRA tensors onto the correct module; a post-load assertion now verifies they match the checkpoint) and **re-scored ColQwen2 with the corrected loader**. The correction is *immaterial* to the headline: fixed-vs-broken gold-page recall@5% is 0.539 vs. 0.534 (+0.005), the top-5% page selections overlap at Jaccard 0.947 (only 34/444 queries change any selected page), and re-reading those changed queries moves fresh-read accuracy by −0.0002 (0.3282 corrected vs. 0.3284 broken); the beat is +0.077 ($\mathrm{LB}_{95} = +0.040$) against the corrected ColQwen2 and +0.077 ($\mathrm{LB}_{95} = +0.039$) against the broken one. The trained projection LoRA does not materially reorder ColQwen2's rankings on this data, so the headline is not loader-inflated. **All headline (and any sealed) numbers use the corrected loader.** Because the correction can only *strengthen* the baseline, the LongDocURL cell—already a matched hold, not a beat—is unaffected, and the page-level appendix history (which predates the fix) inherits the same immateriality.

Separately, the program adopted a **sealed-test discipline**: the 80-document pin is development-only, and a fresh 73/45-document set (433/244 queries; zero overlap) is sealed for a single one-touch final evaluation, which has not run; no number in this paper comes from it.

5 Results

5.1 Reversibility Is Real

At $b = 5\%$ on 903 complete queries (80 documents/cell), the fresh page-level selector scores 0.385/0.258 (LongDocURL/MMLongBench-Doc) vs. 0.128/0.095 for the *identical scorer committed once* and 0.278/0.154 for the gold-informed static. Both pre-registered contrasts clear on both cells and pooled (Table 2); Figure 5 shows every executed isolation interval.

Figure 5 (appendix) shows the tested axis schematically together with every executed isolation interval. Three robustness checks. (i) *Independent extension*: because the screen pin is embedded in the confirmatory pin, we re-ran the statistics on the 50 new documents per cell only ($n = 582$ queries): fresh − commit-once +0.216 $[+0.159, +0.269]$; fresh − static +0.108 $[+0.059, +0.150]$. (ii) *Clean-cache rerun*: the contrast survives the integrity fix of §4.4 (+0.264, $\mathrm{LB}_{95} = +0.187$). (iii) *Scorer generality*: the same fresh-over-stale collapse holds for a retrieval scorer—ColQwen2 fresh 0.423 vs. stale 0.178 (query-micro; equal-cell 0.420 vs. 0.177)—so reversibility is a property of the multi-query regime, not of our scorer. A pre-registered "commitment-pressure" mechanism gradient *failed* its formal interaction test (both intervals include zero) and is retained as exploratory only; we make no mechanism-gradient claim.

5.2 MMLongBench-Doc: The Beat, and Its Audit

The result. On the 80-document dev pin (444 queries decoded; contrasts on the 440 with cached references), PAQ-KV@5% scores 0.4042 vs. ColQwen2-fresh 0.3282: +0.077, $\mathrm{LB}_{95} = +0.040$—the pre-frozen per-cell bar clears with margin (Table 3, Figure 3). ColQwen2 here is loaded with its trained retrieval-projection LoRA correctly applied; §4.4 documents a loader defect we found and corrected, and shows the correction is immaterial to this beat. PAQ-KV attends to a nominal 5% (realized ≈5.9–6.0%) of the context; at that budget it is indistinguishable from full context (+0.009 $[-0.022, +0.040]$) and **not statistically distinguishable from** the gold-page oracle (−0.026 $[-0.059, +0.008]$)—the interval permits material inferiority, so this is parity within noise, not a demonstrated equality.

Correction of an earlier over-claim. An earlier internal draft said PAQ-KV "beats the gold-page oracle" here; the audit showed that to be a small-sample artifact (30-document +0.014 $[-0.042, +0.075]$ vs. 80-document −0.026). The honest statement is only that PAQ-KV, at its ∼5–6% budget, is not statistically distinguishable from the oracle (which reads the exact annotated gold pages, and was pre-registered as *not* a strict ceiling).

The mechanism audit. The beat survives every mechanistic adversarial check available on the **30-document screen** (document-cluster CIs; Figure 3, bottom; these are screen values, distinct from the 80-document headline above):

- **query-dependent**: PAQ-KV − random-block selection = +0.329 $[+0.238, +0.418]$;
- **query-specific**: PAQ-KV − wrong-question selection (select with a *different*, disjoint-gold query's blocks; decode the true question) = +0.698 $[+0.603, +0.784]$—the collapse *supports query-specificity* (a query-invariant or leakage shortcut would not collapse under a wrong question; it does not by itself rule out all leakage);
- **reversible**: PAQ-KV − commit-once = +0.379 $[+0.273, +0.480]$;
- **granularity**: below-page − page-level selection at matched budget = +0.054 $[+0.011, +0.102]$—the granularity itself, not just query-awareness, carries accuracy on this cell.

Caveats, kept visible. The audit contrasts re-analyze 30-document screen data (the beat itself is at 80 documents; its sample-size risk was retired by the top-up). The screen's budget-curve monotonicity gate failed on this cell (0.398 at 10% vs. 0.402 at 5%: selection quality, not budget, moves it; disclosed), and everything here is development-pin evidence pending the sealed one-touch.

Contrast at b=5%	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
fresh − same-scorer commit-once	$+0.257$ $[+0.184, +0.330]$	$+0.163$ $[+0.126, +0.200]$	$+0.210$ $[+0.168, +0.250]$
fresh − gold-informed static	$+0.108$ $[+0.045, +0.164]$	$+0.104$ $[+0.063, +0.144]$	$+0.106$ $[+0.067, +0.141]$
Verdict	REVERSIBILITY ISOLATED (all $\mathrm{LB}_{95} > 0$, both cells and pooled)		

Table 2: Confirmatory isolation at b=5% on 903 complete queries (accuracy; document-cluster paired bootstrap; rows report $\Delta\,[\mathrm{LB}_{95}, \mathrm{UB}_{95}]$). The two-comparator gate was frozen before the run.

	LongDocURL	MMLongBench-Doc
PAQ-KV@5% (ours)	**0.5315**	**0.4042**
ColQwen2-fresh (matched budget)	0.5124	0.3282
full-context (reference)	0.5456	0.3965
gold-page oracle (reference; not strict)	0.6092	0.4314
PAQ-KV − ColQwen2-fresh	$+0.019$ $[-0.019, +0.059]$	$+0.077$ $[+0.040, +0.115]$
PAQ-KV − full-context	-0.014 (n.s.)	$+0.009$ $[-0.022, +0.040]$
PAQ-KV − gold-page oracle	—	-0.026 $[-0.059, +0.008]$
Queries (decoded / paired)	463 / 463	444 / 440
Verdict (pre-frozen bar: per-cell $\mathrm{LB}_{95} > 0$)	HOLD	**BEAT HOLDS**

Table 3: PAQ-KV at the 80-document development pin, nominal 5% budget (all values on the same 80-document dev cohort; document-cluster paired bootstrap; contrast rows $\Delta\,[\mathrm{LB}_{95}, \mathrm{UB}_{95}]$). All four accuracy rows are the verified 80-document dev-pin means—no 30-document screen values are mixed in. On LongDocURL PAQ-KV is *below* full-context (-0.014, not significant); on MMLongBench-Doc it is indistinguishable from full-context. MMLongBench-Doc decodes 444 queries; the paired contrasts against the cached ColQwen2-fresh, full-context, and gold-oracle rows are computed on the 440 queries with cached references (the 4 $\mathrm{mi\_phone}$ rows lack cached references).

Additional matched-budget baselines on the beat cell. Three further comparisons harden the MMLongBench-Doc result, all at the same reader and document-cluster paired bootstrap on the 80-document pin (verdicts released with the code). *(i) Lexical retrieval (BM25).* A BM25 page retriever over per-page text (the benchmark's own text channel where present, tesseract OCR elsewhere; every page covered, no all-zero-score query) scores 0.2109 at matched 5% budget; PAQ-KV beats it by $+0.193$ ($\mathrm{LB}_{95} = +0.152$; 444 queries)—the lexical lower-anchor is reported and beaten, not omitted. *(ii) A training-free CAPS-style expert-head control.* A *page-level* selector using the reader's own cross-validated expert (layer, head) attention—the technique CAPS (Zhu et al. 2026) calibrates, here used training-free—is the strongest page-level attention baseline; it scores 0.3561, and below-page PAQ-KV beats it by $+0.048$ ($\mathrm{LB}_{95} = +0.009$; 444 queries; worst-case missingness imputation leaves the interval unchanged). This isolates the below-page contribution over page-level attention selection at full cohort; the margin is real but thin, and the beat over the *trained retriever* remains the stronger claim. *(iii) Budget robustness.* The beat is not a 5%-only artifact: PAQ-KV at 2.5%, 5%, and 10% all exceed ColQwen2@5% ($+0.103$, $+0.077$, $+0.104$; the 2.5/10% points are on the 157-query screen subset and the 5% point on the full 440), so although PAQ-KV's own budget curve is mildly non-monotone here (above), every budget clears the retriever. A full-pin ColQwen2 re-read at 2.5/10% is future work.

5.3 LongDocURL: Matched, Not Beaten

On the 80-document dev pin (463 queries), PAQ-KV@5% scores 0.5315 vs. ColQwen2-fresh 0.5124: $+0.019$, $\mathrm{LB}_{95} = -0.019$—a positive point estimate that does **not** clear the pre-frozen bar; on this cell PAQ-KV also sits below full context (-0.014, not significant). The mechanism checks (query dependence, reversibility) clear here too; what is missing is the margin. We characterize the shortfall in §6 (a re-analysis of the executed rows suggests the cell is largely reader-conversion-bound rather than selection-bound) and treat it as a bounding observation on where below-page selection helps, not a second headline.

5.4 Negative Ablations, at Full Prominence

Head-mixture transfer gate (the pre-LoRA gate that stopped the B lane): STOP. The B lane aimed to remove benchmark supervision from selection by learning it off-benchmark, ultimately via a toggleable LoRA on the reader's scoring projections. Its *cheapest* pre-frozen check came first, and no LoRA adapter was ever trained: we fit a logistic **per-head mixture** over all $36\times32 = 1{,}152$ (layer, head) question→page attention channels, contrastively, on off-benchmark provenance-checked data (every training page perceptually hashed against all 40,442 benchmark page images), with every design choice selected on external development data. The transfer gate asked whether the externally trained mixture transfers better than a *single* externally selected head—and it does not: mixture gold-page recall $0.652/0.594$ vs. $0.676/0.616$ for the single head; pooled mixture − single $= -0.023$ $[-0.037, -0.008]$ (907 queries).

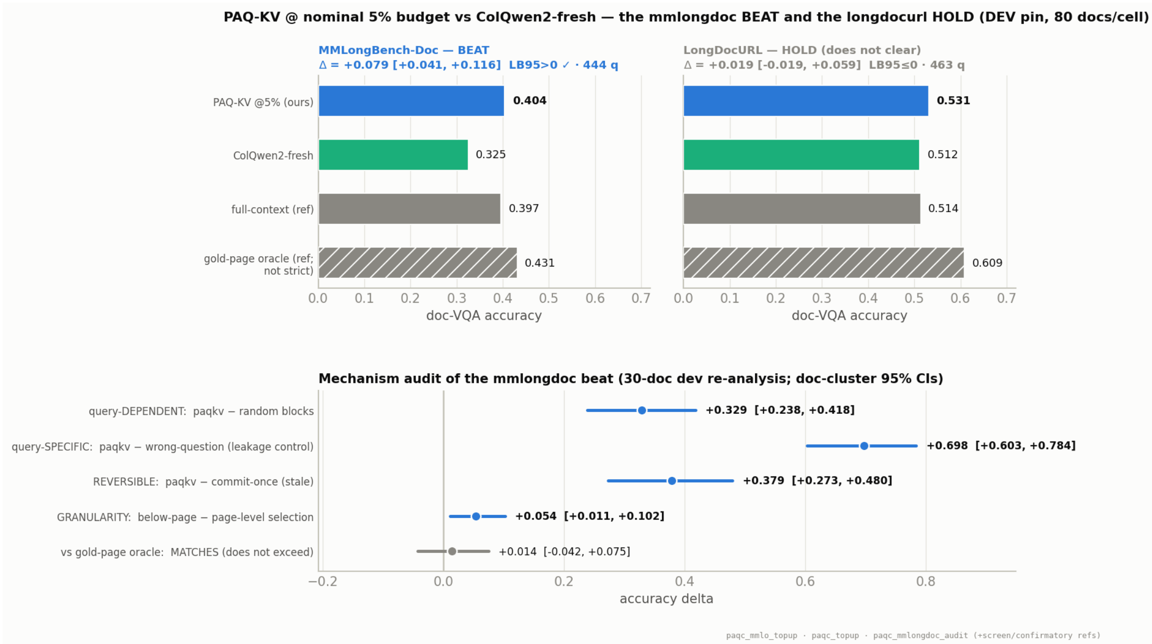

Figure 3: The results panel. **Top:** PAQ-KV@5% vs. ColQwen2-fresh with the full-context and gold-page-oracle references, per cell (80-document dev pin), with document-cluster CIs: the MMLongBench-Doc beat clears ($\mathrm{LB}_{95} = +0.040$); LongDocURL holds ($\mathrm{LB}_{95} = -0.019$). **Bottom:** the mechanism audit of the beat on the **30-document screen** (document-cluster CIs): selection is query-dependent, query-specific (a wrong-question control collapses accuracy, supporting query-specificity), reversible, and below-page $>$ page-level; against the gold-page oracle PAQ-KV is not statistically distinguishable—the CI straddles zero, and we do not claim to exceed it.

The mixture overfits its training distribution, so the lane was closed at this gate. The byproduct is real: the best zero-benchmark-contact head, **L21.H18**, coincides with the expert layer found independently by benchmark-supervised cross-validation—evidence the expert head is a property of the model, not of the benchmarks.

Sharp expert head as block scorer: NO-GO. Swapping PAQ-KV's diffuse mean-attention block scorer (V0) for the transfer-validated sharp head L21.H18 (V1) failed all three pre-frozen conditions: on LongDocURL, V1 nudges the point estimate ($0.5315 \rightarrow 0.536$; $\mathrm{V1} - \mathrm{ColQwen2} = +0.024$ $[-0.012, +0.066]$) but does not separate from V0 ($\mathrm{LB}_{95}(\mathrm{V1} - \mathrm{V0}) = -0.011$, n.s.); on MMLongBench-Doc it *hurts* ($0.402 \rightarrow 0.381$ on the screen documents). The lesson generalizes a recurring theme of this program: **recall-optimized selection is not accuracy-optimized selection**—a sharper, retrieval-style scorer concentrates budget on what looks relevant to a ranking objective and loses the contextual slack the diffuse scorer keeps.

5.5 Why Below Page: The Page-Level History

The program's page-level lineage is retained in full (Appendix B) because it motivates the method. A coarse-preview page router with a full-resolution re-read *loses* to ColQwen2-fresh by $\approx$10 accuracy points at matched page budget (pooled -0.099 $[-0.129, -0.072]$); upgrading its scorer to a cross-validated expert attention head—CAPS's technique (Zhu et al. 2026)—recovers the entire gap to *parity* ($+0.006$/$+0.028$ per cell; pooled $\mathrm{LB}_{95} = -0.000$), not past it, under a pre-registered non-inferiority protocol with an adversarial missingness analysis that erased an initially rosier complete-case edge. Page-level attention selection is published (CAPS), and page-level accuracy saturates at retriever parity; both facts pushed the program below page granularity, where the beat of §5.2 lives.

5.6 Systems Accounting: Simplicity, Not Speed

We measured the deployment profile of the true PAQ-KV path—below-page masked decode over the *retained* full-document KV cache—against ColQwen2-fresh and full context on an 8-document / 31-query systems subset of the pin (single RTX PRO 6000; frozen configuration; Table 4). This measurement corrects an earlier internal accounting that timed a different page-routing variant; it does **not** support a speed claim, and we make none.

Per online query, PAQ-KV costs 1.04 s (0.19 s selection $+$ 0.85 s masked decode) against ColQwen2's 0.46 s: PAQ-KV is $\approx 2.25\times$ *slower* than the trained retriever and never beats it on latency. It is also far heavier in resident state—the retained cache is ≈ 8.15 GiB/document against ColQwen2's ≈ 11 MiB multi-vector index ($\approx 730\times$), and peak VRAM is 44.5 vs 21.8 GiB. Against *full context*, however, PAQ-KV is $\approx 11\times$ faster per query (1.04 vs 11.6 s); because the single prefill amortizes across a document's queries, PAQ-KV's cumulative cost undercuts full-context re-reading after only $N \approx 1.07$

	trained params	per-doc state	online lat./q	peak VRAM
PAQ-KV (ours)	0	8.15 GiB	1.04 s	44.5 GiB
ColQwen2-fresh	2.246 B	11 MiB	0.46 s	21.8 GiB
full-context	0	—	11.59 s	44.6 GiB

Table 4: Deployment profile on an 8-document / 31-query systems subset (frozen configuration, single GPU). PAQ-KV is $\approx 2.25\times$ slower and far heavier in resident state than ColQwen2, and $\approx 11\times$ faster than full context (cold-start amortized after $N \approx 1.07$ queries/document). The win is 0 trained-retriever parameters, no index, one model, and reversible query-time selection—*not* speed.

queries per document (all our documents have more).

The honest efficiency statement is therefore about *simplicity and deployment shape*, not speed. At matched accuracy-beating quality (§5.2), PAQ-KV needs **no separately trained retriever checkpoint** (0 extra trained parameters vs. ColQwen2's 2.246 B), **no stored or maintained vector index**, and **one deployed model** in which retrieval and reading share a single representation; and its query-time selection is **reversible**, so the next question re-selects from the same intact cache rather than being served a committed subset. Those are the properties we claim; the latency and memory cost is a real trade the table states plainly. A fuller table (cold/warm start, co-resident vs. sequential retriever, queries-per-document break-even curves) remains future work.

6 Limitations

- **Development-pin evidence on a reused pin (adaptive-reuse caveat).** Every PAQ-KV number comes from the 80-document development pin, which has been *reused* across the program's screening, auditing, and gating; preregistered gates limit but do not eliminate the resulting adaptive-analysis/multiple-comparisons risk, so all effects here are development-grade until confirmed out-of-sample. The sealed, disjoint 73/45-document set is reserved for a single one-touch final evaluation that has not run; its MMLongBench cell (244 queries / 45 documents) sits below the program's 250-query power floor, so the sealed confirmation needs the effect to hold at slightly reduced power.
- **LongDocURL appears conversion-bound (a bounding observation, not a proven law).** A re-analysis of the executed rows *suggests* the shortfall there is largely past selection: of the 463 rows, $\approx 39\%$ fail under both PAQ-KV and the gold-page oracle (zero even when handed the native-resolution gold pages), only $\approx 10\%$ are oracle-right/PAQ-KV-wrong (the selection-capturable pool), and 2.2% are PAQ-KV-right/oracle-wrong—so the oracle is not a strict ceiling and "no selection method can rescue these" is an observation on this dataset, not a law. Native resolution already saturates the prefill on 79/80 documents, and large documents are a flat, low-leverage half. Among pure-selection levers only a knife-edge oracle-chooser union clears ($LB_{95} = +0.003$, undeployable); only a PAQ-KV$\cup$ColQwen2 hybrid clears comfortably ($LB_{95} = +0.061$) and we exclude it as not a standalone method. We therefore offer the dispersed-vs-conversion-bound split as a *hypothesis* from two datasets.
- **One cell of two.** The beat is scoped to the dispersed-evidence regime; LongDocURL is a matched hold, not a win.
- **Single reader.** All results are a Qwen3-VL-8B case study; generalization to other VLMs is untested.
- **Budget semantics.** The PAQ-KV budget is a nominal 5% decode-attention budget (realized $\approx$5.9–6.0%) with block K/V computed under full-document prefill context—Quest/SnapKV claim semantics, not a matched-token prefill comparison.
- **Audit scale.** The MMLongBench-Doc mechanism-audit contrasts re-analyze 30-document *screen* data (the beat itself is at 80 documents; the audit's sample-size risk was retired by the top-up), and are labelled as screen values throughout.
- **No speed advantage (measured, not deferred).** §5.6 measures the deployment profile on a systems subset: PAQ-KV is $\approx 2.25\times$ *slower* per query and far heavier in resident state ($\approx 730\times$) than ColQwen2, and only $\approx 11\times$ faster than full context. The efficiency argument is simplicity and deployment shape (no separately trained retriever, no index, one model, reversible query-time selection), *not* throughput; the mask route is the accuracy path and physically gathering the cache for memory savings is future work.
- **Base-model ceiling.** 35.1% of confirmatory queries are oracle-unanswerable, deflating absolute numbers for every arm; the query-blind eviction arms are disclosed proxies we claim nothing against.
- **Protocol residues.** The commit-once aggregate uses the first $M = \min(3, k)$ queries in frozen arrival order (order-permutation and M-sensitivity checks remain to be run; the order-free static partially covers this), and the page-level confirmatory run predates the cache fix (§4.4)—its isolation *contrasts* were verified robust, but its absolute fresh-arm levels carry ≈ 0.04 inflation on the 24-document check.

7 Conclusion

PAQ-KV is an architecture that turns a frozen VLM's own KV cache into a queryable, reversible evidence store: prefill once, then per query re-select below-page blocks and decode under a restricted-attention 4D mask over the intact cache, with no gather and M-RoPE preserved. Reversibility is the property this design unlocks, and it holds across scorers and granularities. On dispersed-evidence long-document VQA the full method beats a state-of-the-art trained visual retriever at matched budget and is statistically indistinguishable from the gold-page oracle at $\sim$6% of context, with the mechanism adversarially audited. Where it does not beat the retriever (LongDocURL) we report the matched result as a limitation and characterize the cell as apparently conversion-bound. The sealed held-out evaluation is the single remaining step for out-of-sample evidence, and the dispersed-vs-conversion-bound split is offered as a hypothesis about where below-page reversible selection helps, to be tested on further datasets.

References

Anonymous. 2025a. AGREE: Attention-Guided Retrieval with Dual Supervision for Visual Document Retrieval. *arXiv preprint arXiv:2511.13415*.

Anonymous. 2025b. FOCUS: Internal MLLM Representations for Efficient Visual Cropping. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2506.21710.

Anonymous. 2025c. Patch-to-Region Relevance Propagation for Evidence Localization in Document Visual Question Answering. *arXiv preprint arXiv:2512.02660*. Snappy.

Anonymous. 2025d. SparseVILA: Decoupling Visual Sparsity for Efficient VLM Inference. *arXiv preprint arXiv:2510.17777*.

Anonymous. 2026a. Doc-V*: Coarse-to-Fine Interactive Multi-Page Document Visual Question Answering. *arXiv preprint arXiv:2604.13731*.

Anonymous. 2026b. Hybrid Retriever Evolution for Multimodal Document Reasoning Agents. *arXiv preprint arXiv:2606.29648*.

Anonymous. 2026c. Kamera: Position-Invariant Multimodal KV Cache Reuse. *arXiv preprint arXiv:2606.23581*.

Chen, L.; Zhao, H.; Liu, T.; Bai, S.; Lin, J.; Zhou, C.; and Chang, B. 2024. An Image is Worth 1/2 Tokens After Layer 2: Plug-and-Play Inference Acceleration for Large Vision-Language Models. In *European Conference on Computer Vision (ECCV)*. ArXiv:2403.06764; FastV.

Chen, S.; Gutierrez, B. J.; and Su, Y. 2025. Attention in Large Language Models Yields Efficient Zero-Shot Re-Rankers. In *International Conference on Learning Representations (ICLR)*. ICR; arXiv:2410.02642.

Cho, J.; Mahata, D.; Irsoy, O.; He, Y.; and Bansal, M. 2024. M3DocRAG: Multi-modal Retrieval is What You Need for Multi-page Multi-document Understanding. *arXiv preprint arXiv:2411.04952*.

Deng, C.; Yuan, J.; Bu, P.; Wang, P.; Li, Z.-Z.; Xu, J.; Li, X.-H.; Gao, Y.; Song, J.; Zheng, B.; and Liu, C.-L. 2025. LongDocURL: a Comprehensive Multimodal Long Document Benchmark Integrating Understanding, Reasoning, and Locating. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*. ArXiv:2412.18424.

Dong, K.; Chang, Y.; Goh, X. D.; Li, D.; Tang, R.; and Liu, Y. 2025. MMDocIR: Benchmarking Multimodal Retrieval for Long Documents. In *Proceedings of EMNLP*. ArXiv:2501.08828.

Faysse, M.; Sibille, H.; Wu, T.; Omrani, B.; Viaud, G.; Hudelot, C.; and Colombo, P. 2025. ColPali: Efficient Document Retrieval with Vision Language Models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2407.01449; ColQwen2 is the Qwen2-VL-based variant.

Kim, J.-H.; Kim, J.; Kwon, S.; Lee, J. W.; Yun, S.; and Song, H. O. 2025. KVzip: Query-Agnostic KV Cache Compression with Context Reconstruction. *arXiv preprint arXiv:2505.23416*.

Li, Y.; Huang, Y.; Yang, B.; Venkitesh, B.; Locatelli, A.; Ye, H.; Cai, T.; Lewis, P.; and Chen, D. 2024. SnapKV: LLM Knows What You are Looking for Before Generation. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2404.14469.

Ma, Y.; Zang, Y.; Chen, L.; Chen, M.; Jiao, Y.; Li, X.; Lu, X.; Liu, Z.; Ma, Y.; Dong, X.; Zhang, P.; Pan, L.; Jiang, Y.-G.; Wang, J.; Cao, Y.; and Sun, A. 2024. MMLongBench-Doc: Benchmarking Long-context Document Understanding with Visualizations. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. ArXiv:2407.01523.

Qwen Team. 2025. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *International Conference on Machine Learning (ICML)*.

Tang, J.; Zhao, Y.; Zhu, K.; Xiao, G.; Kasikci, B.; and Han, S. 2024. Quest: Query-Aware Sparsity for Efficient Long-Context LLM Inference. In *International Conference on Machine Learning (ICML)*. ArXiv:2406.10774.

Tito, R.; Karatzas, D.; and Valveny, E. 2024. Self-Attention Based Page Scoring for Multi-Page Document Visual Question Answering. *arXiv preprint arXiv:2404.19024*.

Wan, Z.; Wu, Z.; Liu, C.; Huang, J.; Zhu, Z.; Jin, P.; Wang, L.; and Yuan, L. 2024. LOOK-M: Look-Once Optimization in KV Cache for Efficient Multimodal Long-Context Inference. In *Findings of the Association for Computational Linguistics: EMNLP*. ArXiv:2406.18139.

Wang, Q.; et al. 2025. VRAG-RL: Empower Vision-Perception-Based RAG for Visually Rich Information Understanding via Iterative Reasoning with Reinforcement Learning. *arXiv preprint arXiv:2505.22019*.

Wu, W.; et al. 2025. Query-Focused Retrieval Heads Improve Long-Context Reasoning and Re-ranking. In *Proceedings of EMNLP*. QRHead/QRRetriever; arXiv:2506.09944.

Yuan, Y.; et al. 2025. RegionRAG: Region-level Retrieval-Augmented Generation for Visually-Rich Documents. *arXiv preprint arXiv:2510.27261*.

Zhang, J.; Khayatkhoei, M.; Chhikara, P.; and Ilievski, F. 2025. MLLMs Know Where to Look: Training-free Perception of Small Visual Details with Multimodal LLMs. In *International Conference on Learning Representations (ICLR)*. ViCrop; arXiv:2502.17422.

Zhang, Z.; Sheng, Y.; Zhou, T.; Chen, T.; Zheng, L.; Cai, R.; Song, Z.; Tian, Y.; Ré, C.; Barrett, C.; Wang, Z.; and Chen, B. 2023. H2O: Heavy-Hitter Oracle for Efficient

Group	Arm	LDU	MMLB
ours (page level)	PAQ (fresh)	**0.386**	**0.258**
	attn_max (alt-agg)	0.337	0.152
isolation	same-scorer commit-once	0.128	0.095
	gold-informed static	0.278	0.154
retrieval	ColQwen2-fresh	0.512	0.328
	ColQwen2-stale	0.216	0.138
	CLIP-fresh	0.155	0.164
reference	SnapKV-fresh	0.298	0.157
	full-context	0.546	0.397
	oracle-evidence	0.609	0.431
query-blind	SnapKV-stale	0.083	0.074
	H2O	0.045	0.039
	FastV (proxy)	0.045	0.031
	LOOK-M ($\to$H2O)	0.045	0.039
	KVzip (proxy)	0.047	0.048
	random	0.068	0.022
	truncation	0.033	0.041

Table 5: Every executed arm, accuracy at b=5%, per cell (LDU = LongDocURL, MMLB = MMLongBench-Doc), on the **903-query confirmatory panel** ($\mathrm{paq\_verdict}$). Query-aware arms marked fresh re-select per query; stale arms commit once per document. Cohort note: this 903-query confirmatory panel is the page-level program with the *uncorrected* ColQwen2 loader (ColQwen2-fresh MMLB 0.328); the below-page headline (Table 3) is a distinct top-up cohort (444 decoded / 440 paired) scored with the *corrected* loader (ColQwen2-fresh MMLB 0.3282, §4.4). The loader correction is immaterial, so the shared reference rows agree to rounding.

Generative Inference of Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2306.14048.

Zhu, J.; Bao, Y.; Chen, Y.; and Zhu, W. 2026. Attention as Selector: Unlocking VLM Attention for Long Document Page Retrieval. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026, Long Papers)*, 24344–24365. The CAPS method.

A Full Executed Panel

Table 5 reports every executed arm at the gate budget on the confirmatory pin. The query-blind eviction family sits at or below random gold-page recall on these cells—the anti-strawman record: beating it is visibly not the claim. Secondary pooled statistics: the fresh page-level selector beats the best family arm (SnapKV-stale) with min-over-family $\mathrm{LB}_{95} = +0.202$; fresh $-$ SnapKV-fresh $= +0.094$ $[+0.069, +0.122]$; fresh $-$ oracle $= -0.199$ $[-0.239, -0.161]$. Gold-page recall at b=5% (LongDocURL / MMLongBench-Doc): fresh page selector $0.494/0.443$; gold-informed static $0.381/0.373$; same-scorer commit-once $0.137/0.148$; ColQwen2-fresh $0.664/0.536$.

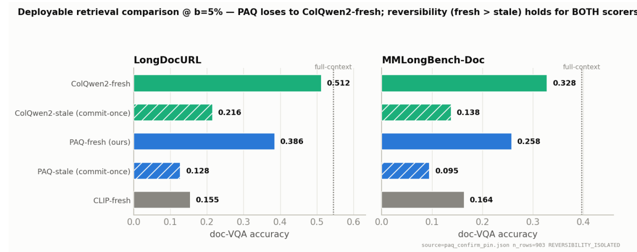

Figure 4: The two honest facts of the page-level history, per cell: (i) ColQwen2-fresh beats the page-level attention router (the $\approx$10-point loss); (ii) *both* scorers collapse when committed once—reversibility is a general property of the multi-query regime, not scorer-specific.

B The Page-Level History in Detail

All numbers in this section are on the **903-query confirmatory cohort** (page-level arms), where ColQwen2-fresh scores $0.512/0.328$ (LDU/MMLB) with the uncorrected loader; this is a distinct cohort from the below-page top-up used in the headline (where the corrected-loader ColQwen2-fresh MMLB is 0.3282, §4.4). The loader correction is immaterial (§4.4), so these page-level parity conclusions are unaffected.

The page router (PAQ v1). Pages are rendered at coarse resolution ($\approx$3.5K visual tokens total, i.e. $\approx$70 tokens/page on a 50-page document) and prefilled once; per query, the question rows' attention (mean over heads, layers, and question tokens) scores pages; the top pages at budget b are re-rendered at full resolution and read. This is page routing, not KV compression, and it claims no memory savings.

The retrieval loss. At $b = 5\%$, ColQwen2-fresh scores $0.512/0.328$ vs. PAQ $0.385/0.258$: pooled -0.099 $[-0.129, -0.072]$—an $\approx$10-point loss that holds on every stratum examined (Figure 4). Gold-page recall is consistent with a coverage contribution: ColQwen2-fresh $0.664/0.536$ vs. PAQ $0.494/0.443$. Both scorers collapse when committed once (ColQwen2 stale $0.216/0.138$), reproducing the reversibility pattern with a second scorer family.

PAQ-T: expert-head recall and the parity read-eval. Scoring with a single expert (layer, head) chosen by held-out-document cross-validation—CAPS's technique—lifts held-out gold-page recall@5% to 0.683 (n=451) / 0.626 (n=425), meeting or exceeding ColQwen2's in-harness $0.664/0.536$; mean aggregation at higher scoring resolution moves almost nothing ($0.494/0.443 \to 0.500/0.452$), so the head, not resolution, is the lever. All selected fold heads sit in layer 21. The pre-registered accuracy read-eval (bars frozen before the run; same reader, render policy, and scorer as the cached baseline rows; only the selected page set differs) lands at **parity**: PAQ-T $0.518/0.357$ vs. ColQwen2-fresh $0.512/0.328$; deltas $+0.006$ $[-0.018, +0.030]$ and $+0.028$ $[+0.001, +0.055]$; pooled $+0.017$ $[-0.000, +0.035]$—non-inferior under the frozen $\mathrm{LB}_{95} > -0.03$ margin on both cells, but not the pre-registered beat (all $\mathrm{LB}_{95} > 0$), and we do not present the razor's-edge per-cell interval as a win.

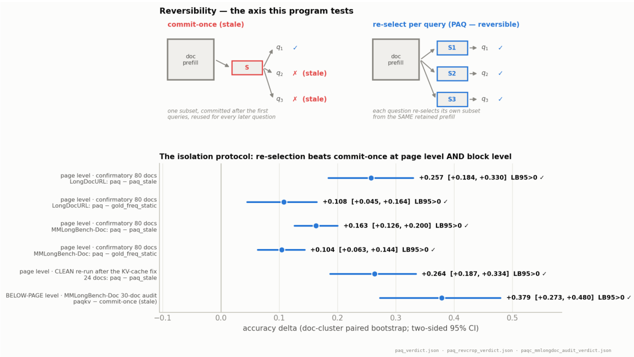

Figure 5: Reversibility—the tested axis. **Top:** a commit-once deployment reuses one subset for every question and goes stale; re-selection re-derives the subset per query from the same retained prefill. **Bottom:** the isolation protocol across granularities: the confirmatory page-level contrasts, the clean re-run after the cache fix, and the block-level PAQ-KV contrast. Every interval clears zero.

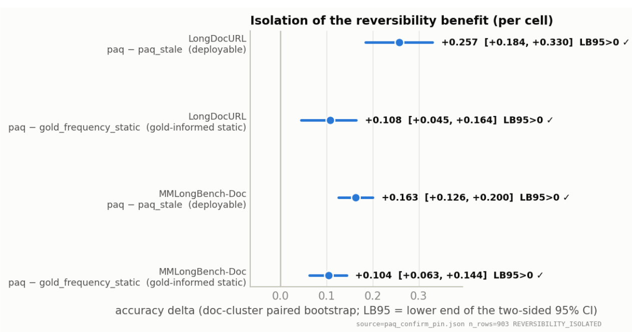

Figure 6: Per-cell forest of the two page-level isolation contrasts at $b = 5\%$. Intervals are document-clustered two-sided 95% CIs; all four exclude zero.

The initially better-looking complete-case verdict (pooled $+0.023$, $\mathrm{LB}_{95} = +0.005$) was *blocked* in review for informative missingness—the $\approx 4\%$ of queries whose high-resolution scoring ran out of memory had cached baseline accuracy 0.62, far above the cell means—and a conservative reduced-resolution recovery of 31/35 missing rows (degrading only the PAQ-T arm, never the baseline) erased the pooled edge. Two caveats carry over: head selection is cross-validated *benchmark-supervised* model selection (not zero-shot; the externally selected L21.H18 of the STOP ablation later removed this caveat from the head's existence, not from its benchmark-tuned use), and the expert-head technique is CAPS's—the residual novelty at page level is only the amortized one-prefill-many-queries architecture.

C Additional Reversibility Detail

Figures 5 and 6 show the reversibility axis schematically and the per-cell forest of the confirmatory isolation contrasts. Additional pre-registered analyses: (i) *answerable subset* (secondary; conditions on a model outcome): on queries where oracle-evidence scores exactly 1 ($n = 407$), fresh 0.583 vs. commit-once 0.182 ($+0.395$ $[+0.324, +0.460]$) and vs. the static 0.382 ($+0.204$ $[+0.135, +0.266]$); (ii) *missingness*: 4 of 907 vision queries are missing due to a per-document timeout; worst-case imputation in either direction leaves the gate verdict unchanged; (iii) *budget curves*: the fresh selector dominates both static comparators at every budget in $\{2.5, 5, 10\}\%$ on both cells; at $b = 10\%$ on LongDocURL it reaches 0.518 vs. full-context 0.546—reading 10% of pages recovers $\approx 95\%$ of full-context accuracy on that cell (not on MMLongBench-Doc: 0.317 vs. 0.397).

Cache-pollution quantification. The in-place cache bug of the main text was quantified before being fixed: on a 6-documents-per-cell blast-radius diagnostic (87 query-rows), the bug inflates gold-page recall by $\approx +0.05$ for a fixed scoring head and shifts selections (mean top-B Jaccard clean vs. polluted 0.82, drifting from 1.00 at q_0 to 0.61 at q_6), as expected for cumulative contamination. The recall gate behind the PAQ-T numbers was regenerated clean before any accuracy read (clean held-out expert recall $0.683/0.626$ vs. polluted $0.674/0.628$), and the clean cross-validation picks a more robust head. Absolute fresh-arm accuracy in the prefix confirmatory carries ≈ 0.04 inflation (clean $-$ polluted -0.041 $[-0.077, -0.003]$ on the 24-document check); the isolation contrasts cancel the bug and stand.

D Reproducibility and Protocol Notes

Pre-registration discipline. Every gate in this program was frozen (bars, comparators, budgets, imputation policy) before its run: the two-comparator isolation gate; the PAQ-T read-eval bars (non-inferiority/beat/inferior margins, per-cell power floors, adversarial missingness imputation); the per-head-mixture transfer gate; the PAQ-KV decode-identity kill criterion, screen gates, per-cell beat bars, and the sharp-scorer (V1) gate. Two early positives did not survive adversarial review and were retracted before this draft: a first isolation gate against the eviction family (invalidated by a weak-static artifact and replaced by the two-comparator gate), and the "beats the gold-page oracle" phrasing (corrected to "matches" at power). The commitment-pressure mechanism hypothesis failed its formal test and is reported as exploratory.

Instrumentation. All significance statements use one instrument: the document-cluster paired bootstrap (10,000 draws, seed 0, percentile intervals; equal-cell pooling), unit-tested alongside the gate and pairing logic. Evaluation pins are sha-256 checksummed and byte-verified before use; the screen pin is a strict prefix of the confirmatory pin, and independent-extension analyses use the 50 new documents per cell only. Figures are generated from the verdict artifacts by a single script; nothing is hand-entered. Per-query, per-arm rows, frozen-bar metadata, verdict files, and the adversarial review trail are retained and will be released with the code.

Sealed one-touch protocol. The 80-document confirmatory pin has been reused across this program and is therefore development-only. The fresh sealed set (73 LongDocURL documents / 433 queries; 45 MMLongBench-Doc documents / 244 queries; zero overlap with the pin) is reserved for a single final evaluation: the baseline will be run on it once, and the only mandate-grade claim this program will make is a scope-appropriate, adversarially imputed, document-clustered $\mathrm{LB}_{95} > 0$ on that sealed set. It has not been evaluated; no number in this paper comes from it.